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Experience

New England Biolabs *Ipswich, MA*
Machine Learning Engineer (Co-op) *May 2026 – Present*

- Built a protein stability ($\Delta\Delta G$) benchmarking harness in Python (∼17.7K lines) evaluating 5 open-source predictors across protein-language-model, task-specific (ProteinMPNN-lineage), and energy-based families on 7 test sets, decoupling prediction from evaluation through an enforced file contract so a new metric re-scores every model with zero GPU hours and a new model takes 3 files
- Surveyed 12+ papers and showed that published Spearman scores are earned on the destabilizing bulk: among stabilizing mutations correlation fell by 0.30–0.51 and was near zero (0.03–0.06) on a 441-protein set, and the best model still required screening 44% of a mutation library to recover the top 3 variants (1.7× over random)
- Designed a per-protein evaluation layer of ∼20 metrics (graded and binary NDCG, recall@k, screening depth, discard-at-recall, AUROC, AUPRC, Spearman) with random baselines beside every ranking metric, eligibility thresholds derived from recall targets, and relevance-mode provenance on every row, covered by 126 unit tests
- Built the data pipeline from Zenodo (Tsuboyama mega-scale, MGnify) to 7 curated test sets of 31 to 552K variant measurements, with leakage checks by protein-level split membership and MMseqs2 coverage-weighted identity that flagged a partial train/test leak (239 of 407 proteins) in the indel set
- Integrated ESMFold and AlphaFold2 (ColabFold) behind a content-addressed structure cache keyed on sequence hash, with atomic, incremental manifest writes so interrupted 20K-sequence folding runs resume, and sharded folding and prediction across GPUs
- Isolated 5 mutually incompatible model environments (e.g. PyTorch 1.12/CUDA 11.3) in per-model conda environments driven by subprocess adapters, and diagnosed silent failures including a sign flip across identical runs of an unseeded energy model and concurrent manifest writes that dropped rows
- Showed predicted stability tracks organism optimal growth temperature across 796 organisms ($R^2 = 0.44$, $\rho = 0.59$), evidence the models capture physical signal beyond their training distribution
- Pitched the harness to an initially skeptical team as a prerequisite for solving stability prediction at scale; adopted as internal tooling by 5 team members, including research scientists
- Built a full-stack protein engineering platform (∼40K lines; FastAPI, Celery/Redis, PostgreSQL, S3/MinIO, Next.js/React, Mol*) that lets bench scientists run 14 tools and 5 models (ThermoMPNN-D, SPURS, ESM3dG, ESM-MSR, ESMFold) through schema-generated forms with linked mutation heatmaps, ranked tables, and 3D structure views
- Designed a plugin architecture where a new tool is one class with zero framework, API, or frontend edits (enforced by test), packaging each model in a pinned Apptainer container that runs as a batch CLI or HTTP server with identical outputs
- Split sequence inputs into chained fold and scoring jobs sharing a content-addressed cache, so jobs hold a GPU only while using it; built a file-lock GPU scheduler and batched ESM3dG inference for a 3.8× faster mutation scan (162 s → 43 s) with identical results
- Verified each predictor's $\Delta\Delta G$ sign convention empirically against ThermoMPNN over 684 shared mutations ($\rho = 0.82$–0.94) and enforced it at each model boundary; backed the platform with 596 automated tests (pytest, vitest) and strict type checking (mypy)
- Added per-residue analysis layers (solvent exposure, secondary structure, Pfam/Swiss-Prot annotation, pockets, interfaces, aggregation) and plain-language input validation so non-computational scientists get failure reasons at submit time, not failed jobs
- Extended in-house variant prediction to cover 3 of the group's 4 protein engineering use cases, adding support for indels, multi-mutants with epistasis, and enzyme discovery, none of which any existing system in the group could score
- Supplied ranked variant predictions to bench scientists across 4 protein engineering campaigns; on the first campaign with experimental results returned, the prior system's ranking would have led the team to the wrong variants, while the expanded pipeline recovered the result

Algoverse *Remote*
AI Researcher *May 2025 – Jan 2026*

- Engineered a model-agnostic Tree of Thoughts (ToT) reasoning framework variant integrating circuit-level interpretability analysis to enable automated identification of stable reasoning paths for pruning, with a modular architecture supporting any language model from Hugging Face and TransformerLens libraries
- Proposed and tested the core research hypothesis that a branch's internal circuit stability is a better signal for pruning reasoning paths than the model's own self-evaluation, the standard approach in ToT
- Led the ToT–interpretability integration in a 6-person research team, contributing 40 of 62 commits and defining the JSON tree format that connects reasoning search to circuit analysis

- Ran per-thought Edge Attribution Patching with Integrated Gradients (EAP-IG) on a 42-layer LLM, extracting top-200-edge circuits from 720K scored edges and using edge-score skewness as a quantitative circuit-stability signal
- Designed the clean/corrupted prompt pairing that makes circuits comparable across reasoning branches, holding a fixed malformed prompt as the corrupted baseline so attribution differences reflect the thought, not the prompt
- Benchmarked reasoning across Game of 24 and GSM8K on GPT-2 and Llama-3.2-3B-Instruct, selecting tasks with verifiable intermediate steps so branch quality could be scored independently of final-answer correctness
- Accelerated research velocity by 75% through an automated GPU cluster deployment pipeline, reducing setup from a 10-step manual process to single-command execution and cutting cloud compute costs by minimizing idle GPU overhead on RunPod
- Delivered an end-to-end ML experimentation pipeline in a 1-week sprint, implementing modified ToT and EAP workflows using PyTorch, TensorFlow, and Hugging Face APIs to enable rapid hypothesis-to-experiment iteration
- Managed large-scale attribution runs across distributed GPU environments, handling multi-gigabyte circuit artifacts and per-variation result organization for a multi-person team
- Authored technical documentation including setup guides, API references, and code documentation, reducing team onboarding time and enabling independent experimentation across distributed research environments

Vellore Institute of Technology *Vellore, India*
Project Associate *Nov 2023 – May 2024*

- Delivered a computer vision system for an industry client in construction, automating identification of reusable site assets (scaffolding, poles, wooden boards, metal sheets) from site imagery
- Curated and annotated a custom dataset of 12,500+ images (2,500 per class × 5 classes) using LabelStudio, designing annotation protocols and implementing quality-control workflows to ensure labeling consistency for multi-class object detection
- Developed an automated Python preprocessing pipeline to handle variability across imaging conditions, implementing background normalization, lighting correction, and augmentation transformations to improve model generalization across uncontrolled outdoor sites
- Built and evaluated ensemble deep learning models (VGG19, Vision Transformers) using PyTorch, conducting systematic hyperparameter tuning and statistical performance analysis to achieve 85%+ accuracy on held-out validation sets
- Presented technical findings and performance metrics to a cross-functional team, communicating model evaluation results and identifying areas for improvement to inform data collection strategies and secure $10K in continued research funding

Skills

Languages: Python, TypeScript, R, SQL, Bash

ML Frameworks: PyTorch, TensorFlow/Keras, Hugging Face, scikit-learn

Deep Learning: Transformers, LLMs, protein language models (ESM), CNNs (VGG19), Vision Transformers (ViT)

Interpretability: TransformerLens, Edge Attribution Patching (EAP-IG), circuit discovery, Sparse Autoencoders (SAEs)

Protein ML & Structure: ESMFold, AlphaFold2 (ColabFold), ThermoMPNN, SPURS, ESM3, MMseqs2, HMMER, DSSP, PyMOL

Genomics & Bioinformatics: Biopython, de Bruijn graph assembly, FastQC, STAR, Kallisto, BWA, DESeq2

LLM Engineering: LangChain, LangGraph, Ollama, RAG, FAISS, sentence-transformers, tool calling, structured outputs (Pydantic)

NLP & Speech: spaCy, NLTK, Microsoft Presidio (de-identification), faster-whisper, docTR (OCR)

Evaluation & Benchmarking: NDCG, recall@k, AUROC/AUPRC, Spearman, random baselines, train/test leakage analysis

Backend & Web: FastAPI, Flask, Next.js, React, Celery, Redis, RabbitMQ, PostgreSQL, MongoDB, SQLAlchemy

Infrastructure & HPC: conda, Apptainer, Docker, Ray, SLURM, Discovery HPC, RunPod, GPU scheduling, S3/MinIO

Testing & Quality: pytest, vitest, mypy (strict), ruff, Git

Data Science: pandas, NumPy, SciPy, networkx, Matplotlib, Jupyter

Tools: LabelStudio, Roboflow, Claude Code (AI-assisted development)

Projects

De Vault: Privacy-Preserving Medical Transcription Systemhttps://github.com/A1pha-Z3r0/De-Vault—Decentralized-VaultJan 2026

- Built an agentic system for an end-to-end medical transcription workflow, orchestrating audio recording, local speech recognition (faster-whisper), de-identification (Microsoft Presidio), and structured data extraction through a fully local LLM pipeline (Gemma 3 4B on Ollama via LangChain)
- Designed prompt-engineered two-stage extraction: a multi-label domain router with confidence scoring feeds 6 domain-specific extractors that generate JSON of patient events, symptoms, diagnoses, and treatment recommendations, validated

against Pydantic schemas with required evidence quotes and cross-field checks
- Implemented a tool-calling framework with specialized functions for medical concept normalization, entity relationship extraction, and data validation, achieving a functional MVP within 1 week

BioRAG: Local Retrieval-Augmented QA for Biomedical Literaturehttps://github.com/A1pha-Z3r0/BioRAGMar 2025
- Built a fully local RAG pipeline for biomedical PDFs so private documents never reach cloud APIs: PyMuPDF loading, 350/50 recursive chunking sized to the embedding model's token limit, all-mpnet-base-v2 embeddings, a FAISS index with top-5 retrieval, and Mistral via Ollama

Lab Notebook Digitization Pipelinehttps://github.com/A1pha-Z3r0/Open-source-Chem-Lab-digitization-pipelineJul – Sep 2025
- Built an event-driven OCR backend for scanned lab notebooks: a FastAPI upload service with SHA-256 deduplication and rollback-safe status tracking in MongoDB GridFS, with Celery over RabbitMQ emitting resumable Redis Streams events to a Ray actor running docTR (LinkNet-ResNet50 + MASTER)

Short-Read Genome Assemblerhttps://github.com/A1pha-Z3r0/short-read-assemblerSep 2025
- Built a de Bruijn graph assembler in Python (networkx, Biopython) using multiplicity-weighted Eulerian traversal per component, with a conservative unitig fallback joined only on unique overlaps, plus per-contig coverage statistics and histograms

Education

Northeastern University *Boston, MA*
Master of Science in Bioinformatics *Expected May 2027*
Relevant Coursework: Algorithms, Genomics in Bioinformatics, Transcriptomics, Bioinformatics Programming, Data Collection, Storage and Retrieval in R

Vellore Institute of Technology *Vellore, India*
Bachelor of Technology in Biotechnology *Sep 2020 – May 2024*
Relevant Coursework: Bioinformatics, Molecular Modelling and Drug Design, Genetic Engineering, Cancer Biology and Informatics